MRR Project - Genes & Breast Cancer - EDA

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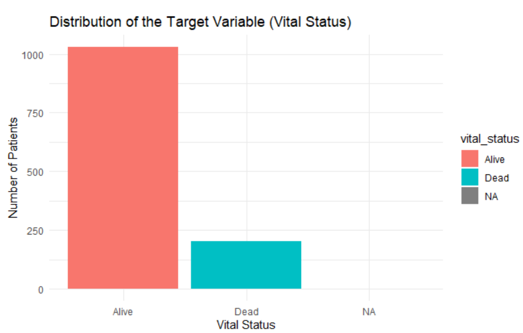
This study uses two complementary dataset : a high-dimensional gene expression matrix and a detailed clinical dataset. Together, they enable us to explore how molecular profiles relate to breast-cancer patient outcomes, particularly vital status, our target variable.

Gene Expression Dataset (GeneX)

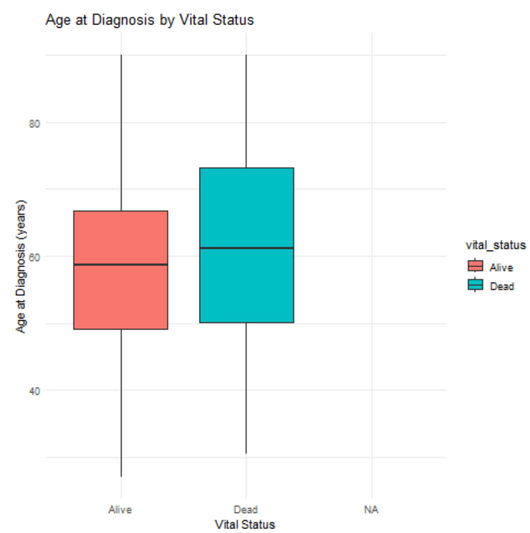
The `GeneX` dataset contains continuous expression values for 5,000 genes measured across 1,231 patients. Patients are represented by rows, and genes by columns. This dataset supports the detection of co-expression patterns, the identification of gene groups, and the analysis of their relationship with survival outcomes.

Clinical Dataset (`clinical_data`)

The `clinical_data` table includes 24 clinical and demographic variables describing the same 1,231 breast-cancer patients. It covers tumor characteristics, patient attributes, treatment information, and follow-up outcomes.



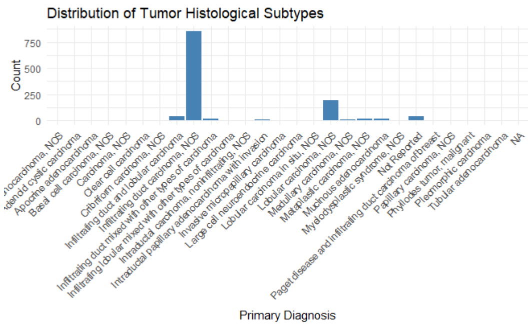
(a) Barplot - vital status distribution



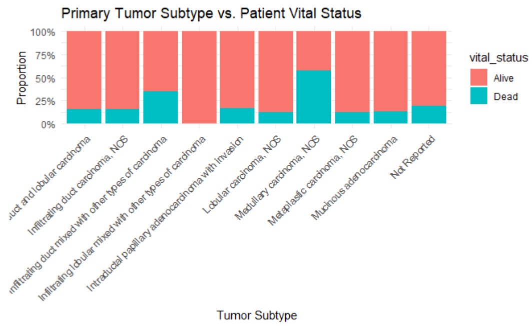
(b) Boxplot - age at diagnosis by vital status

Figure 1: Exploration of patient vital status distribution and its relationship with age at diagnosis.

As shown in Figure 1a, the cohort includes 1,029 alive patients and 201 deceased patients. Moreover, Figure 1b shows that the mean age at diagnosis is lower among alive patients (58.5 years) compared with those who died (61.6 years), suggesting an age-related difference in survival outcomes.

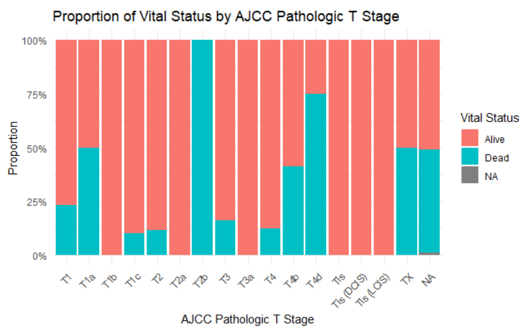


(a) Barplot - tumor distribution

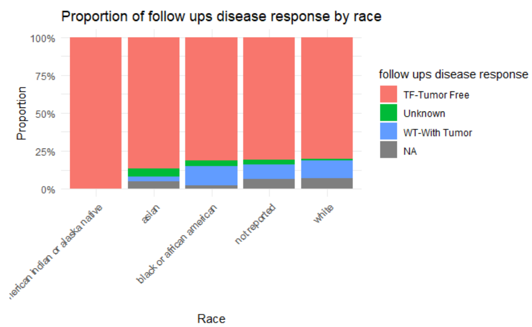


(b) Barplot - Tumor subtype vs vital status

Figure 2: Overall tumor type distribution and subtype differences according to vital status.



(a) Barplot - Vital Status by T stage



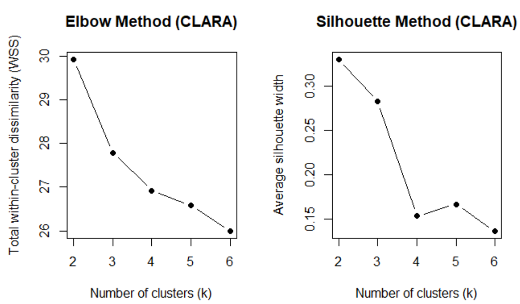
(b) Barplot - Follow ups disease response by race

Figure 3: Vital status across AJCC T-stages and follow-up disease response by race.

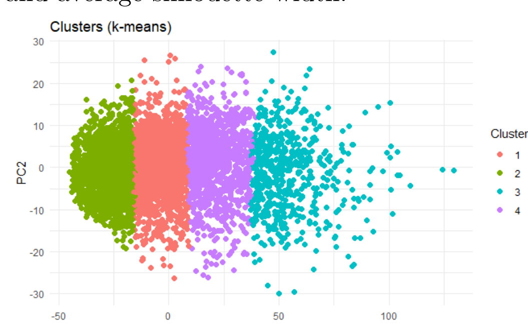
Figure 2a shows that infiltrating duct carcinoma, NOS (857 cases) is the predominant tumor type, followed by lobular carcinoma, NOS (195 cases). Survival differences across subtypes are notable (Figure 2b): medullary carcinoma has the highest mortality (57%), whereas infiltrating lobular mixed carcinoma shows 100% survival. Vital status also varies strongly with AJCC T-stage (Figure 3a). Early-stage tumors such as T1c show low mortality (10%), while more advanced stages perform considerably worse, with T2b at 100% mortality and T4d at 75%. Finally, follow-up disease responses differ across races (Figure 3b). Asian patients exhibit the lowest rate of persistent disease (3.3%), followed by White patients (11.8%) and Black or African American patients (12.7%), suggesting disparities in post-treatment outcomes.

Clustering : grouping genes together

We cluster genes using **k-means** to simplify visualisation, assuming genes with related expression patterns group together. Because genes must be treated as observations, the dataset is transposed so genes form the rows. To select the **optimal number of clusters *k***, we use methods (PAM-CLARA), and assess cluster quality using two complementary metrics: within-cluster sum of squares (WSS) and average silhouette width.



(a) Graphs of methods to determine k.



(b) Visualisation of genes clustering ($k = 4$) using PCA

Figure 4: Determination of the optimal number of clusters and visualization of genes clustering.

The high silhouette width and the flattening elbow curve (Figure 4a) indicate that $k = 4$ is optimal (Figure 4b).